Computer Vision Basics

AI Learning Hub - Comprehensive Course Guide

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Image Processing Fundamentals

Digital images are represented as matrices of pixel values.

Key Operations:
- Color Spaces: RGB, HSV, Grayscale conversion.
- Filtering: Gaussian blur, median filter for noise reduction.
- Edge Detection: Sobel operator, Canny edge detector.
- Morphological Operations: Erosion, dilation, opening, closing.
- Histogram Equalization: Enhancing image contrast.

Libraries: OpenCV provides comprehensive tools for image manipulation. Images are loaded as NumPy arrays, enabling fast mathematical operations.

Object Detection and Segmentation

Object Detection identifies and localizes objects in images.

Evolution of architectures:
- R-CNN family: R-CNN, Fast R-CNN, Faster R-CNN with Region Proposal Networks.
- Single-shot detectors: YOLO (You Only Look Once), SSD.
- YOLO divides image into grid, predicts bounding boxes and class probabilities.

Image Segmentation:
- Semantic Segmentation: Classify each pixel (FCN, U-Net, DeepLab).
- Instance Segmentation: Detect and segment each object instance (Mask R-CNN).
- Panoptic Segmentation: Combines semantic and instance segmentation.

Transfer Learning for Vision

Transfer learning uses pre-trained models as starting points.

Common pre-trained models (trained on ImageNet):
- ResNet: Skip connections solve vanishing gradient problem.
- EfficientNet: Compound scaling of depth, width, resolution.
- Vision Transformer (ViT): Applies transformer architecture to image patches.

Fine-tuning strategies:
1. Feature Extraction: Freeze all layers, train only new classifier head.
2. Fine-tuning: Unfreeze some or all layers, train with small learning rate.
3. Progressive unfreezing: Gradually unfreeze layers from top to bottom.

Data Augmentation: Random crops, flips, rotations, color jitter, Mixup, CutMix.